

Zomato Restaurant Analytics Project Report

1. Introduction

The objective of this project is to analyze Zomato restaurant data and generate business insights that help understand customer preferences, restaurant performance, cuisine popularity, pricing patterns, and geographic trends.

The project follows an end-to-end analytics approach using Excel, Python, MySQL, SQL, and Power BI.

2. Tools Used

Tool	Usage
Excel	Initial Data Cleaning
Python (Pandas)	Data Transformation
Jupyter Notebook	Data Processing
MySQL	Database Creation
SQL	Business Querying
Power BI	Dashboard Creation

3. Data Pipeline

Raw Dataset

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Excel Cleaning

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Python (Pandas)

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MySQL Database

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SQL Analysis

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Power BI Dashboard

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Business Insights

4. Data Cleaning in Excel

Performed the following operations:

- Removed duplicate records.
 - Handled missing values.
 - Corrected inconsistent entries.
 - Standardized column names.
 - Improved overall data quality.
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5. Data Transformation using Python

Using Pandas library:

- Imported Excel data.
 - Converted data types.
 - Cleaned and transformed columns.
 - Created analysis-ready dataset.
 - Exported processed data to MySQL.
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6. Database Design

Database:

zomato_db

Table:

zomato_data

Data was loaded into MySQL for business analysis.

7. SQL Analysis

Business questions answered:

Restaurant Analysis

- Which restaurants have the highest ratings?
- Which cities contain the largest number of restaurants?

Cuisine Analysis

- Which cuisines are most popular?
- Which cuisines receive higher ratings?

Cost Analysis

- Average cost by city.
- Cost distribution across restaurants.

Geographic Analysis

- Country-wise restaurant distribution.
- City-wise restaurant concentration.

Service Analysis

- Online delivery availability.
 - Table booking analysis.
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8. Power BI Dashboard

Dashboard components include:

KPI Cards

- Total Restaurants
- Average Rating
- Average Cost
- Number of Cities

- Number of Countries

Charts

- Restaurant Count by City
 - Cuisine Distribution
 - Rating Analysis
 - Country Distribution
 - Price Range Analysis
 - Online Delivery Analysis
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9. Key Findings

Restaurant Performance

- High-rated restaurants are concentrated in major cities.

Cuisine Trends

- Certain cuisines dominate customer preferences.

Geographic Insights

- A limited number of countries contribute the majority of restaurants.

Pricing Insights

- Mid-priced restaurants represent the largest segment.

Service Insights

- Online delivery adoption varies significantly across locations.
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10. Conclusion

This project demonstrates a complete Data Analytics lifecycle:

Data Cleaning

(Excel)

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Data Transformation

(Python + Pandas)



Data Storage

(MySQL)



Data Analysis

(SQL)



Dashboard Development

(Power BI)



Business Insights

The project showcases practical skills in:

- Excel
- Python
- Pandas
- SQL
- MySQL
- Power BI
- Data Cleaning
- Data Visualization
- Business Analysis

Author

Dr. Monika Yadav

Business Analyst | Data Analyst

SQL • Power BI • Excel • Python • Tableau

Turning Data into Insights.